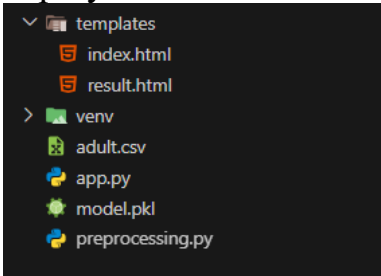


**Course Title: AI Foundations, Machine Learning, Deep Learning, NLP,
and Computer Vision**

Modules Covered upto:21-07-2025 To 25-07-2025 (20 Hours)

Modules	Title	Hours
	Login Steps configuration of Data Bricks • Data Analysis in Databricks • Analysing Data in Databricks • Create a Sample Data Frame • Summary Statistics • Data Visualization in Databricks • Filtering & Sorting • Load Data from CSV • Create a Spark Session • View Schema.	20hours
	Introduction Algorithms and its Implementation. • Linear Regression Algorithms • logical Regression Algorithms • Random Forest Classifier Algorithms • Decision Tress Classifier Algorithms • Simple linear regression metrics R2-score • Multiple linear regression metrics R2-score • Random Number Generation process using NumPy and pandas for data manipulation process. • PCG64 Random number generation • Finding model performance summary Hyperparameter tuning is the process for Machine learning • How does the Hyperparameter Tuning Operate or Function • Applications of Hyperparameter Tuning • Advantages of Hyperparameter Tuning • Hyperparameter tuning affect model training time Scikit-learn: Machine Learning in Python • Install Scikit-learn	

	• Pipeline • Make of pipeline • Simple Inputer • Column Transformer • Preprocessing of data • OneHotEncoder • StandardScaler • RandomForestRegressor • Sklearn model selection using GridSearchCV	
	Sklearn Model selection and Train and Test data splitting process using (X_train, y_train). Finding Accuracy score, Classification Report, Confusion Matrix using Sklearn. • How to Identify columns. • How to Define models. • How to Store results into Dictionary. • How Training and Evaluation with Pipeline process.	
	Implementation of Randomized SearchCV and RandomForestClassifier • How to define combined models with proper pipelines. • Finally, plotting graphs of ROC and LOC curve	
	Installation of PyCharm IDE process. • Creating Project steps and configurations • Installation Related package to create and run application	

	Created of Movie model creation and implementation in to web application (Flask). using DecisionTreeClassifier ML algorithm • Creating and downloading Movie datasets • Dropping of interest data columns for the input and output for model training. • Fitting the data into the model • Predict the model with proper input and finding its accuracy prediction. What is pickle and its uses? • How to install and implement pickle in both Jupyter notebook and PyCharm • Creating Dump using PyCharm. • Generating .mkl file for the model.	
	Deploy Machine Learning Model using Flask • Installing procedure for machine learning deploy in flask.  • Steps for Creating Application structure to the flask implementation. • Creating the app.py file the code implementation. • Execution and deploy the application. • Different between. pkl and h5 file	
	Created Taxi application model using RandomForestClassifier: • Training and splitting the data • Finding the Accuracy score • Model implementation • Save the model in .pkl file • Run the app	

	Created the Login process the RoboFlow application in the browser • What is RoboFlow and its application • Object detection process and steps explain, • Downloaded cat and dog image detection done. • Final Test and deploy processes done. Note: On Friday Some Student have facing issue in creating the model creation and implementing into the application so their doubts were cleared and explained the procedure and individual attention has given on that day for clearing all their queries. **All the Teams have Completed the Movie Prediction Application model creation and uploaded on their cloud Drives. Team Leaders have nominated to Monitoring the team members completion and uploading on their drives and reporting to me the progress of completion or any issues faced by the particular students.	
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